

ADARSH SINGH

Noida, Uttar Pradesh, India — Open to relocation anywhere

+91-6386247822 — singhadadarsh9240@gmail.com

LinkedIn — GitHub — LeetCode — Portfolio

Education

JSS Academy of Technical Education (AKTU)

Bachelor of Technology in Information Technology (CGPA: 7.54)

Oct 2023 – Oct 2027

Noida, India

Work Experience

HELLO LIVE NA Pvt. Ltd.

Software Development Engineer Intern (Backend & AI)

May 2026 – Present

Remote, India

- Architected a high-frequency algorithmic trading engine using **Node.js, Python, and WebSockets** to ingest real-time tick data from multiple exchange APIs, processing **10,000+ messages/second** with sub-**5ms** latency.
- Engineered a predictive execution module leveraging **LLM APIs and time-series forecasting** for live market sentiment analysis, optimizing trade entry/exit parameters and driving a **22% increase** in backtested portfolio ROI.
- Designed a fault-tolerant backend utilizing **PostgreSQL and Redis** for state management and high-speed order-book caching, cutting database latency by **40%** and ensuring **99.99% uptime** during market volatility.

Projects

OckhamGrid | Distributed GitOps & AI Code Optimizer | Go, Rust, Python, Docker Jun 2026 – Present

- Architected a distributed polyglot microservice mesh (Go Sandbox, Rust AST Engine, FastAPI) to autonomously detect $O(N^2)$ algorithmic bottlenecks, slashing enterprise cloud computing costs by up to **40%** via CPU cycle optimization.
- Orchestrated a Zero-Touch GitOps pipeline integrating OpenAI LLMs for deterministic code refactoring; achieved sub-**400ms** AI inference latency and executed automated GitHub PR submissions, reducing manual code-review overhead by **85%**.
- Engineered fault-tolerant Dockerized containers with isolated Goroutine keep-alive mechanisms and graceful network degradation; mitigated TCP port collisions to secure **99.99%** pipeline uptime across distributed cloud nodes.

Vyzrox | Zero-Latency Edge AI Engine | Rust, WebAssembly, WebGPU, Web Workers Mar 2026 – May 2026

- Engineered a groundbreaking 100% serverless ambient AI suite, compiling Rust to WebAssembly (WASM) to execute a local neural core entirely in-browser, achieving an unprecedented **1ms** inference latency with 0 external API calls.
- Spearheaded hardware-accelerated local inference by distributing 0.5B+ parameter neural computations across background Web Workers and WebGPU, yielding sustained generation speeds of **45+ tokens/second** while maintaining flawless 60 FPS UI rendering.
- Pioneered a strict Zero-Trust privacy architecture guaranteeing 100% data sovereignty; eliminated 100% of cloud hosting overheads and mitigated all network-based data-breach vectors for local multimodal analysis.

Q-Ecosystem | Secure B2C SaaS Platform | Next.js 14, Turborepo, Supabase, Clerk Feb 2026 – Apr 2026

- Architected a Turborepo monorepo of 4 interconnected micro-apps (**Qrento, Q-Store, Q-Delivery, Q-Partner**) to manage premium lifestyle asset rentals, achieving 60% code reuse and accelerating CI/CD by 45%.
- Provisioned scalable PostgreSQL schemas with strict Row Level Security (RLS) for absolute transactional data isolation, enforcing RBAC and secure JWT handshakes via Clerk API Gateway.
- Mitigated bot abuse via Sliding Window rate-limiting; streamlined route groups for Zero-CLS, achieving sub-**80ms** p99 response latency and sub-**200ms** First Contentful Paint (FCP).

Achievements

Competitive Programming (LeetCode): Peak Rating: 1868 (Knight), solved 600+ algorithmic problems across arrays, graphs, and DP with a 300-day max streak.

Certifications

Oracle: Cloud Infrastructure 2025 Certified AI Foundations Associate

Oct 2025

Google Cloud: Introduction to Generative AI

Sep 2025

DeepLearning.AI: Building AI Voice Agents for Production

Feb 2025

Technical Skills

Core CS: Data Structures & Algorithms, OOP, System Design, DBMS, Operating Systems, Computer Networks

Languages: C++, Go, Rust, JavaScript (ES6+), TypeScript, Python, SQL

Frontend & Architecture: React.js, Next.js 14/15, Tailwind CSS, Turborepo (Micro-frontends), WebAssembly

Backend & APIs: Node.js, Express.js, FastAPI, RESTful APIs, Microservices, WebSockets, Pub/Sub

Databases & Caching: PostgreSQL, MongoDB, Redis, Supabase, Mongoose ODM, Query Optimization

AI / LLM: OpenAI API, Gemini API, Prompt Engineering, Edge AI (WebGPU), LLMops, AST Analysis

DevOps, Auth & Tools: Git, Docker, CI/CD (GitHub Actions), Linux, Vercel, Render, Clerk, JWT